Data Activity 2: Creation of a frequency table, calculation of percentages, and creation of a binary variable.

```
> # Upload file
> covid19cases <- read.csv(file.choose())
>
> # Data set structure
> str(covid19cases)
'data.frame': 270 obs. of  7 variables:
 $ Sno                     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                    : chr  "30-01-2020" "31-01-2020" "01-02-2020" "02-02-2020" ...
 $ State.UnionTerritory    : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths                  : int  0 0 0 0 0 0 0 0 0 0 ...
>
>
> # Corrects date format as it initially appears as a character format
> covid19cases$Date <- as.Date(covid19cases$Date, format = "%d-%m-%Y")
>
>
> # Task 1 Create a frequency table to count how many daily reports showed recoveries
versus no recoveries during this period. Use the table() command.
> table(covid19cases$Cured > 0)

FALSE  TRUE
  215    55
>
>
> # Task 2 Calculate percentages to understand what proportion of daily state reports
included recovery cases.
> prop.table(table(covid19cases$Cured > 0)) * 100

   FALSE     TRUE
79.62963 20.37037
>
>
> # Task 3 Create a binary variable called has_recovery that indicates whether any
recoveries (cured cases) were reported.
> covid19cases$has_recovery<- ifelse(covid19cases$Cured > 0, 1, 0)
>
>
> covid19cases$has_recovery #Creation of new variable has_recovery
  [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0
0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0
0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
[125] 1 0 0 0 0 0 0 0 0 0 0 0 1 0 1 1 0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 0 1 0 1 0 0 1 0 0 1 0 0
1 0 0 0 1 0 1 0 0 1 0 0 1 0 0 1 0 0 0 1 0 1 0 0 1 0 0 1 0 0 1 0 0 0 0 1 0 1 0 0 1 0 0 0 0
1 0 0 1 0 0 0 0 1 1 1 0 0 0 1 0 0 0 0 1 0 0 1 1 0 0 0 0 1 1 1 0 0 0 1 0 0
[249] 0 0 1 0 0 0 1 1 0 0 0 0 0 1 1 1 0 0 0 1 0 0
>
>
> # 1 Additional scripts to make the presentation of the data readable.
>
> #Creation of new frequency table using binary variable where 0 = No recoveries, 1 =
Recoveries reported
> table(covid19cases$has_recovery)

  0   1
215  55
>
> # Creation of revised percentages calculation in line with using binary variable
> prop.table(table(covid19cases$has_recovery)) * 100

       0        1
79.62963 20.37037
>
```

```
> # 2 Additional scripts to make the presentation of the data readable.
>
> #Scripts to give the output clear readable labels where 0 = No recoveries, 1 =
Recoveries reported
> table(factor(covid19cases$has_recovery,
+              levels = c(0, 1),
+              labels = c("No Recoveries", "Recoveries Reported")))

      No Recoveries Recoveries Reported
                215                  55
>
> #Scripts to give the output clear readable labels where 0 = No recoveries, 1 =
Recoveries reported and calculate percentages
> prop.table(table(factor(covid19cases$has_recovery,
+              levels = c(0, 1),
+              labels = c("No Recoveries", "Recoveries Reported")))) * 100

      No Recoveries Recoveries Reported
           79.62963            20.37037
>
> #Scripts to give the output clear readable labels where 0 = No recoveries, 1 =
Recoveries reported, calculate percentages and round to 2 decimal places
> round(prop.table(table(factor(covid19cases$has_recovery,
+                         levels = c(0, 1),
+                         labels = c("No Recoveries", "Recoveries Reported")))) * 100, 2)

      No Recoveries Recoveries Reported
              79.63               20.37
```